

Yuze Fu

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 Website |  LinkedIn |  Github

RESEARCH INTERESTS

Numerical Methods for Partial Differential Equations, Physics-based Computer Graphics

EDUCATION

- **University of Toronto** Sep 2023 - June 2027 (expected)
Honours Bachelor of Science - Mathematics and Computer Science Toronto, Ontario, Canada
 - GPA: 3.99/4.00
 - Math Courses: Abstract Algebra, Numerical Methods for PDEs, Number Theory, Graph Theory, Analysis
 - CS Courses: Machine Learning, Operating System, Database, Human-computer Interaction
- **High School Affiliated to Fudan University** Sep 2020 - June 2023
High School Diploma Shanghai, China
 - Studied Math, Physics, Chemistry, Geography
 - Member of the school's competitive programming team for the Informatics Olympiad
 - Member of the school's soccer team, playing against other high schools in Shanghai

RESEARCH

- **Adaptive Particle Splitting for Fluid Simulation** May 2026 - Present
Supervised by Eitan Grinspun and Jonathan Panuelos Dynamic Graphics Project Lab
 - Developing adaptive particle splitting for a meshless, moving-Voronoi fluid solver (incompressible and compressible), adding resolution where the flow needs it without remeshing the whole domain — in C++ as Houdini plugins.
 - Designed a conservative refinement rule that divides a particle's mass, momentum, and energy among its children by cell volume, keeping the simulation exactly conservative and removing the pressure and density artifacts that naive splitting causes.
 - Developed a conservative remap that corrects density errors introduced when the mesh re-tessellates around a split, transferring mass between refined cells and their neighbors by geometric cell overlap — making refinement effectively invisible on uniform flow and verified by conservation, positivity, and entropy tests.
- **Automated Misinformation Detection & Claim Verification Pipeline** Jun 2025 - Aug 2025
Research Assistant Rotman School of Management
 - Developed a two-phase NLP pipeline using SBERT and LLMs to classify social media misinformation.
 - Applied cosine similarity and embedding-based filtering to align thousands of tweet-article pairs.
 - Leveraged DeepSeek API to classify claim relationships as identical, contradictory, or unrelated.
 - Optimized system accuracy via iterative threshold tuning and incremental data processing strategies.

TEACHING

- **Teaching Assistant for CSCB09 (System Programming)** Winter 2026
- **Teaching Assistant for CSCB36 (Introduction to the Theory of Computation)** Fall 2025
- **Teaching Assistant for MATA22 (Linear Algebra I for Mathematical Science)** Winter 2025
- **Teaching Assistant for CSCA67 (Discrete Mathematics)** Fall 2024

HONORS AND AWARDS

- **Department of Computer Science Award** 2026
University of Toronto \$12000
 - 16-Weeks Undergraduate Student Research Award funded by the Department of Computer Science.
- **University of Toronto Scholar** 2024
University of Toronto \$1500
 - University-wide in-course scholarships recognized for most outstanding academic achievements.
- **Dean's List** 2024, 2025, 2026
University of Toronto Scarborough
- **Second Prize in the National Olympiad in Informatics (NOIP2020)** 2020
China Computer Federation

PROJECTS

- **Classification of Finitely Generated Modules over PIDs** Mar 2026 - Apr 2026
Abstract Algebra Essay
 - Classified finitely generated modules over PIDs to unify the structure of abelian groups and operators.
 - Formulated the bijection between $F[x]$ -modules and linear maps to prove operator structural identities.
 - Constructed Rational and Jordan Canonical Forms using Invariant Factor and Elementary Divisor forms.
 - Concluded with a review of the theoretical and computational trade-offs between different matrix canonical forms.
- **Numerical Analysis of Finite Difference Schemes for Hyperbolic PDEs** Feb 2026 - Apr 2026
Numerical Methods for PDEs Projects
 - Implemented 8 finite difference schemes using vectorized NumPy operations, optimizing performance for advection-based simulations.
 - Evaluated the stability and order of convergence for standard schemes (e.g. Lax-Wendroff, Leapfrog) under varying CFL conditions.
 - Developed SBP-SAT boundary treatments for high-order Runge-Kutta methods, ensuring stable and consistent convergence up to the 4th order.
 - Analyzed theoretical vs. experimental trade-offs between ghost cell extrapolation and SBP-SAT operators for boundary stability.
- **NSI Forecasting: Machine-Learning-Driven Neighborhood Safety Index** Nov 2025 - Dec 2025
Machine Learning Project
 - Engineered a custom Neighborhood Safety Index (NSI) based on the Canadian Crime Severity Index to quantify urban safety levels.
 - Evaluated Ridge, KNN, and FNN models to compare baseline linear accuracy against non-linear neural approximations.
 - Developed an LSTM network achieving a leading R^2 of 0.8295 by modeling complex 12-month temporal sequences.
 - Optimized performance through grid search tuning of architecture depth, activation functions, and k-neighbors.

COMMUNITY INVOLVEMENT

- **UTSC Accessibility Services** Sep 2024 - Present
Volunteer Notetaker
 - Produced comprehensive lecture and tutorial notes to support students with diverse accessibility needs.
 - Courses: Probability (STAB52), Computation Theory (CSCB36), Group Theory (MATC01), Numerical Methods for PDEs (MATD10)
- **UTSC Summer School** Jul 2025 - Aug 2025
Teaching Assistant
 - Helped Professors to hold seminars about computer science topics for high school students.
 - Answered students' questions and provided information about university lives.
- **The Association of Mathematical and Computer Science Student (AMACSS)** Sep 2024 - Apr 2025
Course Representative
 - Developed review materials and facilitated pre-exam seminars to improve peer academic performance.
 - Courses: Discrete Math (CSCA67), Introduction to Computer Science (CSCA08)
- **UTSC Google Developer Group (GDG)** Oct 2024 - Apr 2025
Academic Director
 - Supported the execution of "Build with AI," an industry-focused event with 500+ registrations and live AI demos.
 - Created an interactive emotion-matching game that successfully increased participant engagement.

SKILLS AND HOBBIES

- **Programming Languages:** C/C++, Python, Assembly, SQL, Shell Script, Java, R, LaTeX, Markdown
- **Frameworks and Libraries:** pandas, NumPy, SciPy, PyTorch, scikit-learn
- **Developer Tools:** Houdini, Docker, PostgreSQL, Git
- **Hobbies:** Soccer, Piano, Swimming, Traveling, Movie, RPG Game